Here is a **simple and student-friendly explanation** of **Conditional Probability**.

Conditional Probability

Conditional probability is the probability of an event happening **when we already know that another event has happened**.

Definition

If **A** and **B** are two events, then the probability of **A occurring given that B has already occurred** is called the **conditional probability** of A given B.

It is written as:

$$P(A|B)$$

where:

- **P(A|B)** = Probability of A occurring given that B has occurred.
- "|" is read as **"given that"**.

Formula

Initially, conditional probability is defined as:

$$P(A|B) = \frac{N(A \cap B)}{N(B)}$$

where:

- **N(A ∩ B)** = Number of outcomes common to both A and B.
- **N(B)** = Total number of outcomes in event B.
- **N(B) ≠ 0**

Derivation of the Formula

Let **N** be the total number of outcomes in the sample space.

Start with:

$$P(A|B) = \frac{N(A \cap B)}{N(B)}$$

Multiply both numerator and denominator by **N**:

$$P(A|B) = \frac{\frac{N(A \cap B)}{N}}{\frac{N(B)}{N}}$$

We know:

$$\frac{N(A \cap B)}{N} = P(A \cap B)$$

and

$$\frac{N(B)}{N} = P(B)$$

Therefore,

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

where

$$P(B) \neq 0$$

Multiplication Rule

Rearranging the above formula,

$$\boxed{P(A \cap B) = P(B) \times P(A|B)}$$

Similarly,

$$\boxed{P(A \cap B) = P(A) \times P(B|A)}$$

where

$$P(A) \neq 0$$

Formula for $P(B|A)$

The probability of **B occurring when A has already occurred** is

$$\boxed{P(B|A) = \frac{P(A \cap B)}{P(A)}}$$

provided

$$P(A) \neq 0$$

Easy Example

Suppose a class has **30 students**.

- 18 students play **Cricket (C)**.
- 12 students play **Football (F)**.
- 8 students play **both Cricket and Football**.

Find the probability that a student plays Cricket **given that** the student plays Football.

Step 1: Use the formula

$$P(C|F) = \frac{P(C \cap F)}{P(F)}$$

Using the number of students,

$$P(C|F) = \frac{8}{12}$$

$$P(C|F) = \frac{2}{3} = 0.667$$

Answer

There is a **66.7% chance** that a student plays Cricket **if we already know that the student plays Football**.

Key Points to Remember

- Conditional probability finds the probability of one event **after another event has already occurred**.
- The formula is:

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

- The multiplication rule is:

$$\boxed{P(A \cap B) = P(B) \times P(A|B)}$$

or

$$\boxed{P(A \cap B) = P(A) \times P(B|A)}$$

- Conditional probability is only defined when the probability of the given event is **not zero**.

Quick Summary

Formula	Meaning
(P(A	B)=\dfrac{P(A\cap B)}{P(B)})
(P(B	A)=\dfrac{P(A\cap B)}{P(A)})
(P(A\cap B)=P(B)\times P(A	B))
(P(A\cap B)=P(A)\times P(B	A))

Memory Tip:
Think of conditional probability as **"restricting the sample space."** Once event **B** has happened, only the outcomes in **B** are considered. Among those outcomes, we calculate the chance that **A** also occurs.

Is this conversation helpful so far?